

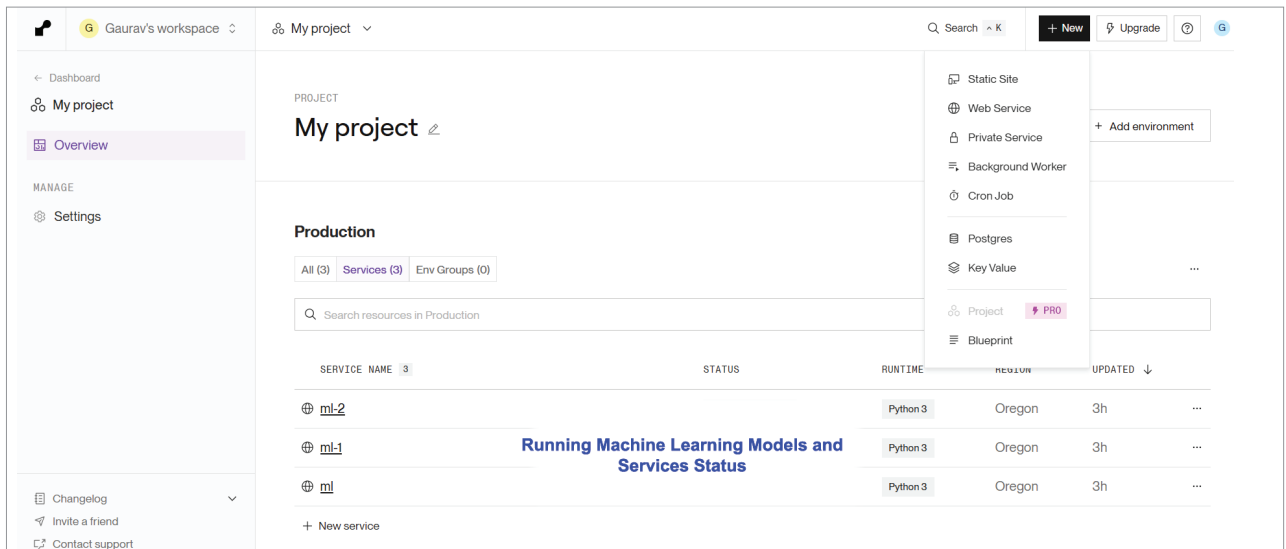


Figure 6: Projects and apps on Render cloud

```

"""Retrieve relevant documents and
generate an answer."""
# Embedding of the question
query_embedding = myretriever.
encode([question], convert_to_
numpy=True)

# Retrieve top document
_, indices = index.search(query_
embedding, k=1)
myretrieved_doc =
documents[indices[0][0]]

# Generate answer with T5
input_text = f"Question: {question}
Context: {retrieved_doc}"
myinput_ids = mytokenizer.
encode(input_text, return_tensors="pt",
truncation=True)
output_ids = mygenerator.
generate(myinput_ids, max_length=50)
answer = mytokenizer.decode(output_
ids[0], skip_special_tokens=True)

return f"Answer: {answer}\n\
nContext: {myretrieved_doc}"

# -----
# Step 4: Gradio Interface
# -----
giface = grgui.Interface(
    fn=myrag_query,
    inputs=grgui.Textbox(label="Enter

```

```

your question"),
    outputs=grgui.
Textbox(label="Generated Answer"),
    title="AI RAG Application",
    description="As a Question, Response
will be displayed from trained data"
)

giface.launch()

```

On Hugging Face Spaces, AI-based applications can be deployed for image generation, text generation, video generation, language translation, speech synthesis, object detection, 3D modelling, optical character recognition (OCR), etc.

Render

URL: <https://render.com>

Render is another powerful and multi-

featured cloud platform for deploying AI-based applications. It has the features to deploy web applications, static sites, Docker containers, and cron jobs required for Industry 4.0 applications. These applications can be uploaded on a GitHub account and then linked to Render for quick deployment.

The platform provides load balanced autoscaling, DDoS protection, infrastructure as code as well as data privacy for industrial applications.

Researchers would do well to develop and deploy their machine learning and deep learning based applications on Hugging Face Spaces, Render or similar platforms so that the web-based interface to their project has better accessibility as well as security of the source code. **END** 🐧

References

- Artificial Intelligence Market (2025 - 2033), <https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-market>
- Render, <https://render.com/>
- Hugging Face Spaces, <https://huggingface.co/spaces>
- Gradio, <https://www.gradio.app/>

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